Proposal Form Additional Validation - PRD

Adult Count, Child Count, Tenure, Height, Weight, ABHA Number and Relationship-Title Validation

1. Purpose

This document defines backend validation requirements for Proposal Form member counts, policy tenure, height, weight, ABHA Number, and relationship-title mapping. The validations must run before the proposal is submitted to Policy Planner. All applicable validation errors should be returned as field-level errors.

2. Validation Summary

Field	Required Rule
adultCount	Whole number; minimum 1 and maximum 4; must match adults[] count
childCount	Whole number; minimum 0 and maximum 4; must match children[] count
selectedTenure	Only 1 year is allowed
height	Numeric value from 30 to 250 cm, inclusive
weight	Numeric value from 5 to 300 kg, inclusive
abhaId	Optional; if supplied, validate as 14-digit ABHA Number
relationship/title	Father must use Mr; Mother must use Mrs

3. Adult Count Validation

Business rule: A proposal can contain a maximum of 4 adult members.

Scenario	Result	Validation Message
adultCount is missing/null	Invalid	adultCount is required
adultCount is not a whole number	Invalid	adultCount must be a whole number
adultCount = 0 or negative	Invalid	adultCount must be between 1 and 4
adultCount = 1, 2, 3 or 4	Valid	-
adultCount > 4	Invalid	adultCount must be between 1 and 4
adultCount does not equal populated adults[] entries	Invalid	adultCount must match the number of adult members
adultCount = 4 and adults[] contains 4 members	Valid	-
adultCount = 4 but adults[] contains 5 members	Invalid	Maximum 4 adult members are allowed

Count comparison must use actual populated adult member records. Null/empty placeholder objects must not be treated as valid adult members.

4. Child Count Validation

Business rule: A proposal can contain 0 to 4 children.

Scenario	Result	Validation Message
childCount is missing/null	Invalid	childCount is required
childCount is not a whole number	Invalid	childCount must be a whole number
childCount < 0	Invalid	childCount must be between 0 and 4
childCount = 0	Valid	-
childCount = 1, 2, 3 or 4	Valid	-
childCount > 4	Invalid	childCount must be between 0 and 4
childCount does not equal populated children[] entries	Invalid	childCount must match the number of child members

childCount = 0 but children[] contains member data	Invalid	childCount must match the number of child members
childCount = 4 and children[] contains 4 members	Valid	-
children[] contains more than 4 members	Invalid	Maximum 4 child members are allowed

5. Count Cross-Validation

adultCount	Actual Adults	childCount	Actual Children	Result
1	1	0	0	Valid
4	4	4	4	Valid
2	3	1	1	Invalid - adult count mismatch
3	3	2	1	Invalid - child count mismatch
5	5	0	0	Invalid - adult maximum exceeded
2	2	5	5	Invalid - child maximum exceeded

6. Tenure Validation

Business rule: Proposal tenure is fixed at 1 year.

Input	Result	Validation Message
Missing/null	Invalid	selectedTenure is required
1	Valid	-
0	Invalid	selectedTenure must be 1 year
2 or greater	Invalid	selectedTenure must be 1 year
Negative value	Invalid	selectedTenure must be 1 year
Non-numeric/unsupported value	Invalid	selectedTenure must be 1 year

The backend must not silently convert an unsupported tenure to 1 year.

7. Height Validation

Business rule: Height must be between 30 cm and 250 cm, inclusive.

Scenario / Input	Result	Validation Message
Missing/null when height is required	Invalid	Height is required
Non-numeric value	Invalid	Height must be a numeric value
Negative or zero	Invalid	Height must be between 30 and 250 cm
29	Invalid	Height must be between 30 and 250 cm
30	Valid - lower boundary	-
175	Valid	-
250	Valid - upper boundary	-
251	Invalid	Height must be between 30 and 250 cm

If height is collected for multiple insured members, validate each populated member independently and return the error against that member's height field.

8. Weight Validation

Business rule: Weight must be between 5 kg and 300 kg, inclusive.

Scenario / Input	Result	Validation Message
Missing/null when weight is required	Invalid	Weight is required

Non-numeric value	Invalid	Weight must be a numeric value
Negative or zero	Invalid	Weight must be between 5 and 300 kg
4	Invalid	Weight must be between 5 and 300 kg
5	Valid - lower boundary	-
70	Valid	-
300	Valid - upper boundary	-
301	Invalid	Weight must be between 5 and 300 kg

Do not silently truncate decimal values. If Policy Planner accepts decimal weight, validate the decimal value against the same 5-300 kg range. If the downstream contract permits only whole numbers, reject decimals explicitly.

9. ABHA Number Validation

The existing JSON field may remain abhaId. For validation purposes it represents the ABHA Number. ABHA is optional. When a value is supplied, it must pass all local format validations below.

Scenario	Result	Validation Message
Null/blank	Valid	-
Exactly 14 numeric digits	Format valid	-
Fewer than 14 digits	Invalid	ABHA Number must contain exactly 14 digits
More than 14 digits	Invalid	ABHA Number must contain exactly 14 digits
Contains alphabetic characters	Invalid	ABHA Number must contain digits only
Contains spaces	Invalid	Invalid ABHA Number format
Contains hyphens	Invalid	Invalid ABHA Number format
Contains other special characters	Invalid	Invalid ABHA Number format

Local format rule: exactly 14 digits. Suggested pattern: ^[0-9]{14}$.

Passing local validation means the ABHA Number is format-valid only. Local validation must not claim that the ABHA exists, is active, or belongs to the proposer/insured member. Such verification requires an authorized ABDM verification flow.

Do not introduce Aadhaar-specific rules (for example, the existing repeated-digit Aadhaar rule) into ABHA validation unless a separate confirmed business/ABDM requirement is provided.

10. Relationship and Title Validation

Business rule: Father must have title Mr. Mother must have title Mrs.

Relationship	Title	Result	Validation Message
father	Mr	Valid	-
father	Mrs	Invalid	Title must be Mr when relationship is father
father	Ms	Invalid	Title must be Mr when relationship is father
father	Missing/null	Invalid	Title must be Mr when relationship is father
mother	Mrs	Valid	-
mother	Mr	Invalid	Title must be Mrs when relationship is mother
mother	Ms	Invalid	Title must be Mrs when relationship is mother
mother	Missing/null	Invalid	Title must be Mrs when relationship is mother

Relationship-title validation is independent of DOB/age validation. Example: Father + Mr can pass the title rule but still fail the existing Father-to-Self minimum 21-year age-gap rule.

11. Duplicate Adult Relationship Validation

Recommended proposal consistency rule: only one member should occupy each unique adult relationship role (Self, Spouse, Father, Mother). This prevents adultCount from being technically valid while the member composition is invalid.

Scenario	Result	Validation Message
One Self	Valid	-
Two Self records	Invalid	Duplicate adult relationship is not allowed
One Spouse	Valid	-
Two Spouse records	Invalid	Duplicate adult relationship is not allowed
Two Father records	Invalid	Duplicate adult relationship is not allowed
Two Mother records	Invalid	Duplicate adult relationship is not allowed

12. Combined Validation Scenarios

Scenario	Result / Expected Error
adultCount=4, exactly 4 valid adults	Valid
adultCount=5	Invalid - adultCount must be between 1 and 4
adultCount=2 but 3 adults supplied	Invalid - adultCount must match the number of adult members
childCount=0 and children[] empty	Valid
childCount=0 but one child supplied	Invalid - childCount must match the number of child members
childCount=4, exactly 4 children	Valid
childCount=5	Invalid - childCount must be between 0 and 4
selectedTenure=1	Valid
selectedTenure=2	Invalid - selectedTenure must be 1 year
height=30 cm	Valid lower boundary
height=250 cm	Valid upper boundary
height=29 or 251 cm	Invalid - Height must be between 30 and 250 cm
weight=5 kg	Valid lower boundary
weight=300 kg	Valid upper boundary
weight=4 or 301 kg	Invalid - Weight must be between 5 and 300 kg
ABHA blank and optional	Valid
ABHA supplied with exactly 14 digits	Format valid
ABHA supplied with 13/15 digits	Invalid - exact 14 digits required
Father + Mr + valid age gap	Valid for these rules
Father + Mrs	Invalid - Title must be Mr when relationship is father
Mother + Mrs + valid age gap	Valid for these rules
Mother + Mr	Invalid - Title must be Mrs when relationship is mother
Father + Mr but Father/Self age gap < 21	Invalid under existing DOB/Age validation

13. Validation Execution Order

1. Validate adultCount and childCount data type/range.
2. Validate adults[] and children[] maximum size.
3. Cross-check adultCount against populated adults[].
4. Cross-check childCount against populated children[].
5. Validate selectedTenure = 1.
6. Validate each applicable member height.
7. Validate each applicable member weight.
8. Validate ABHA Number only when it is supplied.
9. Validate relationship-title mapping for Father and Mother.
10. Run existing DOB, age, relationship-gap, and child-sequence validations.
11. Collect all detected field-level errors and return HTTP 400 if any validation fails.
12. Call Policy Planner only when all BoltPe validations pass.

14. Validation Message Catalogue

Field	Validation Message
adultCount	adultCount is required
adultCount	adultCount must be a whole number
adultCount	adultCount must be between 1 and 4
adultCount	adultCount must match the number of adult members
adults	Maximum 4 adult members are allowed
childCount	childCount is required
childCount	childCount must be a whole number
childCount	childCount must be between 0 and 4
childCount	childCount must match the number of child members
children	Maximum 4 child members are allowed
selectedTenure	selectedTenure is required
selectedTenure	selectedTenure must be 1 year
height	Height is required
height	Height must be a numeric value
height	Height must be between 30 and 250 cm
weight	Weight is required
weight	Weight must be a numeric value
weight	Weight must be between 5 and 300 kg
abhaId	ABHA Number must contain exactly 14 digits
abhaId	ABHA Number must contain digits only
abhaId	Invalid ABHA Number format
title	Title must be Mr when relationship is father
title	Title must be Mrs when relationship is mother
relationship	Duplicate adult relationship is not allowed

15. API Error Behaviour

When one or more validations fail, BoltPe should return HTTP 400 Bad Request and include all detected field-level errors. Policy Planner must not be called for a request that fails BoltPe validation.

Example:

```
{
 "status": 400,
 "error": "Bad Request",
 "message": "Proposal validation failed",
 "fieldErrors": {
  "adultCount": "adultCount must match the number of adult members",
  "selectedTenure": "selectedTenure must be 1 year",
  "adults[1].height": "Height must be between 30 and 250 cm",
  "adults[1].title": "Title must be Mr when relationship is father"
 }
}
```

16. Final Acceptance Criteria

- No proposal can contain more than 4 adults.
- No proposal can contain more than 4 children.
- Declared adult/child counts must match actual populated member arrays.
- Only 1-year tenure is accepted.
- Height boundaries 30 cm and 250 cm are accepted; values outside are rejected.
- Weight boundaries 5 kg and 300 kg are accepted; values outside are rejected.
- ABHA is optional. Null/blank is valid; when supplied, the ABHA Number must pass format validation.
- ABHA local validation does not represent ABDM identity/existence verification.

- Father accepts Mr only; Mother accepts Mrs only.
- These validations do not replace existing DOB/age and relationship-gap validations.
- All applicable errors are returned together before any Policy Planner API call.